TECHNICAL DELIVERY WORK SAMPLE

AI Assisted Imaging Referral Intake

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I would pilot AI-assisted intake in one imaging clinic's fax queue. The system drafts a referral record and points staff to the relevant text in the original fax. An intake coordinator reviews each draft and approves every referral that moves to scheduling.

Pilot scope and assumptions

The fictional clinic starts with knee MRI referrals. For this exercise, a referral needs a verified patient match, the ordering clinic and a way to reach it, the requested exam, and the side stated on the current order. Other exam types continue through the clinic's usual intake process. Before building, I would confirm the intake checklist, who answers clarification requests and whether scheduling can look up a referral by ID.

Workflow and responsibilities

Stage	Action and owner
Receive	The intake system assigns a referral ID and keeps the original fax. The coordinator identifies referrals eligible for the pilot.
Draft and validate	AI proposes field values with page references and short quotes. Intake rules flag missing or conflicting information.
Review	The coordinator checks the source, confirms the patient match, corrects fields and either approves the handoff, requests clarification or sends the case to manual intake.
Hand off	The scheduling queue accepts at most one approved record per referral ID. The scheduling team handles appointment booking.

The intake lead owns the review rules and exception queue. Engineering owns the fax and scheduling integrations. Clinical priority stays with the clinic's existing review process.

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Referral Review and Exceptions

A synthetic referral with a missing field

REF-DEMO-104 | Page 1 | Patient PAT-DEMO-018
Referrer Demo Clinic 07 | Reply route ROUTE-DEMO-07
Requested exam: MRI knee
Clinical history: Prior right knee injury

Field	Simulated AI draft	Coordinator review
Exam	MRI knee, from the order on page 1	Accepted as written.
Side	Right, citing “Prior right knee injury” on page 1	Cleared the proposed value. The side is missing from the current order.

The coordinator moves the referral to Needs clarification and contacts the ordering clinic. The clinic replies, “Current order is for an MRI of the left knee.” The coordinator attaches the reply, verifies the corrected side and approves the handoff.

I would use this failure as a regression test. The side must be supported by text in the current order or a clarification recorded by staff. A quote from the history section leaves the field unresolved. If the source sections are ambiguous, the case goes to manual review.

Exception handoffs

Condition	Next action
Unreadable fax or uncertain match	The coordinator uses manual intake. The case stays out of scheduling until the source and patient match are verified.
Scheduling timeout	The referral stays Pending handoff. The system checks scheduling for the referral ID. If no record exists, it retries with the same ID. Engineering investigates repeat failures.

The intake record links the original fax, AI suggestions, staff edits and scheduling receipt. It records who approved the handoff and when.

Pilot Tests and Launch Criteria

Acceptance tests

Synthetic case	Expected result
Complete order and verified match	Coordinator approval creates one scheduling item. A receipt marks the handoff complete.
Complete order awaiting approval	The record stays in intake. Scheduling receives nothing.
Side missing from the order	The case goes to Needs clarification, even when the history mentions a side.
Unreadable fax or ambiguous match	The case goes to manual intake. Scheduling stays blocked.
Same fax receipt delivered twice	The intake system keeps one referral record and one review task.
Scheduling receives the record but the reply times out	The system finds the item by referral ID and records the receipt. One scheduling item remains.

Baseline and proposed targets

Time manual and AI-assisted review on matched sets of 30 synthetic referrals with the same mix of complete and exception cases. Compare median active staff minutes from opening the fax to the first decision, including time spent checking and correcting fields. Track the share of known incomplete cases the system flags before staff review.

For this exercise, I'd propose at least 20% less median review time, all planted missing-side cases flagged, and zero wrong-patient, unapproved or duplicate scheduling handoffs. A move from 6.0 to 4.5 minutes would be a 25% reduction. Those figures illustrate the calculation.

Launch decision and fallback

Run the rehearsal in a test scheduling queue. Launch in one queue once all tests pass, the targets are met and both the intake lead and engineering sign off. During rollout, the intake lead reviews corrections and engineering monitors pending transfers. After an incorrect or unapproved handoff, switch to manual intake and reconcile pending transfers before resuming.